

PROJECT TITLE

CGPA Calculator

- **Project Description**

The Student CGPA Calculator is a simple C++ program that helps students calculate their GPA and CGPA based on the grades and credit hours obtained in different courses. The program accepts course details from the user, calculates total credits and grade points, and displays the final GPA/CGPA. This project demonstrates the use of arrays, loops, user input, arithmetic calculations, and output formatting in C++.

- **Objectives**

To take input for multiple courses.

To store grades and credit hours.

To calculate total grade points.

To calculate GPA and CGPA.

To display course-wise details and final results.

- **Algorithm**

1. Start the program.

2. Enter the number of courses.

3. For each course:

Enter grade point.

Enter credit hours.

Calculate grade point $\times$ credit hours.

4. Calculate total credits.

5. Calculate total grade points.

6. Compute CGPA using:

$$\text{CGPA} = \text{Total Grade Points} / \text{Total Credits}$$

7. Display all course details.

8. Display the final CGPA.

9. End the program.

- **C++ Program**

```
#include <iostream>
```

```
#include <iomanip>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int n;
```

```
    float grade[20], credit[20];
```

```
    float totalCredits = 0, totalGradePoints = 0, cgpa;
```

```
    cout << "Enter Number of Courses: ";
```

```
    cin >> n;
```

```
    for(int i = 0; i < n; i++)
```

```
    {
```

```
        cout << "\nCourse " << i + 1 << endl;
```

```
        cout << "Enter Grade Point: ";
```

```
        cin >> grade[i];
```

```
        cout << "Enter Credit Hours: ";
```

```

    cin >> credit[i];

    totalGradePoints += grade[i] * credit[i];
    totalCredits += credit[i];
}

cgpa = totalGradePoints / totalCredits;

cout << "\n----- Course Details ----- \n";
cout << "Course\tGrade\tCredits\n";

for(int i = 0; i < n; i++)
{
    cout << i + 1 << "\t"
        << grade[i] << "\t"
        << credit[i] << endl;
}

cout << "\nTotal Credits: " << totalCredits;
cout << "\nTotal Grade Points: " << totalGradePoints;
cout << "\nFinal CGPA: " << fixed << setprecision(2) << cgpa;

return 0;
}

```

- **Sample Input**

Enter Number of Courses: 4

Course 1

Enter Grade Point: 9

Enter Credit Hours: 4

Course 2

Enter Grade Point: 8

Enter Credit Hours: 3

Course 3

Enter Grade Point: 10

Enter Credit Hours: 4

Course 4

Enter Grade Point: 7

Enter Credit Hours: 2

Sample Output

----- Course Details -----

Course Grade Credits

1 9 4

2 8 3

3 10 4

4 7 2

Total Credits: 13

Total Grade Points: 114

Final CGPA: 8.77

Formula Used

Grade Points = Grade $\times$ Credit Hours

CGPA = Total Grade Points $\div$ Total Credits

Applications

Academic performance tracking.

College and university result analysis.

Semester GPA and overall CGPA calculation.

Student record management systems.

- **Conclusion**

The Student CGPA Calculator efficiently calculates a student's academic performance by considering grades and credit hours of all courses. It is a simple and useful application that demonstrates fundamental C++ programming concepts such as loops, arrays, calculations, and user interaction.